

ENGINEERING COLLEGE PLACEMENT & RECRUITMENT INTELLIGENCE

A DATA-DRIVEN STUDY

Insights. Analysis. Decisions that Drive Better Placements.



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PROJECT DURATION
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ABSTRACT

Engineering colleges in India vary considerably in placement performance, employability outcomes, internship participation and average starting salary. Recruiters, placement cells, students and institutional management often lack a consolidated, data-driven view that connects these outcomes to branch, tier, ownership, category, state and skill-related factors. This project addresses that gap by building an analytical dataset of 22,386 records across 2,813 unique colleges, 33 states, 7 engineering branches, 26 categories, 7 ownership types and 4 tiers, and by developing a Power BI decision-support framework on top of it.

The central business problem examined is why engineering college placements are low in aggregate, which institutional and branch-level factors are associated with stronger placement outcomes, and how recruiters can identify colleges and branches that demonstrate stronger placement, employability, internship participation, salary outcomes and recruiter readiness.

The dataset was cleaned and standardized through a structured methodology covering column identification, data-type validation, missing-value inspection, duplicate inspection, state standardization, category and ownership cleaning, branch standardization, numerical validation and analytical grouping, using College_Name, State_Clean and Branch_Name as the analytical uniqueness key. The final dataset contains zero missing values and zero duplicate analytical records.

Descriptive statistics show an overall mean placement percentage of approximately 63.13%, mean average salary of approximately 5.78 LPA, mean internship participation of approximately 54.69%, and mean employability indicator of approximately 61.72%. Branch-level analysis indicates that Computer Science Engineering records the highest average placement (74.36%) and salary (7.16 LPA), while Civil Engineering records the lowest average placement (49.34%) and salary (4.28 LPA). Tier-level analysis indicates that Tier 1 institutions show the strongest average performance across all four quantitative indicators, while Tier 3 and Tier 4 show relatively similar average performance.

A project-defined Recruiter Readiness framework was developed using Placement Percentage, Employability Indicator and Internship Participation to classify institutions as Recruiter Ready, Moderately Ready or Needs Improvement. This framework, together with descriptive fields on curriculum subjects, recruiter demand, industry skills, and skill gaps, was translated into a set of Power BI dashboards supporting recruiter-oriented college and branch shortlisting.

The project concludes that placement outcomes are associated with a multi-factor combination of branch, tier, employability and internship participation rather than any single determinant, and that a combined, multi-metric, dashboard-driven approach is more informative for recruiters and institutions than any single indicator considered in isolation. All findings are presented as observed associations rather than causal claims.

Keywords: Engineering Placements, Employability, Internship Participation, Recruiter Readiness, Skill Gap Analysis, Power BI, Exploratory Data Analysis, Branch-wise Analysis, College Tier, Decision Support System.

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INTRODUCTION

Background

Placement outcomes are among the most closely tracked performance measures for engineering colleges in India. Students, parents, recruiters and accreditation bodies routinely use placement percentage and average salary as proxies for institutional quality. However, placement performance is not uniform: it varies substantially by engineering branch, institutional tier, ownership type, category and state. Despite this variation being widely acknowledged informally, structured, data-driven analysis that connects placement outcomes to employability, internship participation, salary and skill-alignment factors at scale is comparatively rare.

This project responds to that gap by constructing a cleaned, analysis-ready dataset of engineering college placement records and building an Exploratory Data Analysis (EDA) and Power BI-based decision-support framework around it.

Problem Statement

The central business problem examined in this project is: why are engineering college placements low, what institutional and branch-level factors are associated with stronger placement outcomes, and how can recruiters identify colleges and branches that demonstrate stronger placement, employability, internship participation, salary outcomes and recruiter readiness?

Need for the Study

Recruiters currently face difficulty in identifying which colleges and branches are genuinely recruiter-ready, since publicly available information is often limited to college reputation, historical brand perception or isolated placement percentage figures without further context. Similarly, placement cells and college management often lack a structured, comparative view of how their institution performs relative to peer institutions across tier, branch, state, category and ownership. Students, in turn, often lack visibility into branch-level and skill-related employability patterns when making academic and career decisions. This project is intended to address these informational gaps using a single, cleaned analytical dataset and an associated Power BI decision-support layer.

Research Questions

1. What is the overall placement performance of engineering colleges in the dataset?
2. How does placement performance vary across engineering branches?
3. How do college tiers differ in placement, salary, internship participation and employability?
4. Is stronger employability associated with stronger placement outcomes?
5. How does internship participation differ across branches and tiers?
6. What patterns are observed in average salary across branches and tiers?
7. What skill gaps are visible between college-taught skills and recruiter-demanded skills?

8. Which institutions demonstrate stronger recruiter readiness under the project's defined framework?
9. How can Power BI support recruiter-oriented college shortlisting?
10. What areas should engineering colleges prioritize to improve placement readiness?

Objectives

- To clean, standardize and validate a large multi-source engineering college placement dataset into a single analytical dataset free of missing values and duplicate analytical records.
- To compute and interpret descriptive statistics for placement percentage, average salary, internship participation and employability indicator.
- To analyze placement, salary, internship and employability patterns across engineering branches and institutional tiers.
- To examine differences across states, institutional categories and ownership types.
- To document recruiter demand, industry skills required, skills taught in colleges, skills demanded by recruiters, missing skill gaps and emerging technology requirements as recorded in the dataset.
- To design a project-defined Recruiter Readiness framework combining placement, employability and internship participation.
- To build Power BI dashboards that operationalize the above analysis for recruiter-oriented and institution-oriented decision support.
- To develop recommendations for colleges, placement cells, recruiters and students grounded strictly in the observed data.

Scope

The scope of this project is limited to the 22,386 analytical records contained in the final dataset, covering 2,813 unique colleges, 33 states, 7 engineering branches, 26 categories, 7 ownership types and 4 tiers. The analysis is descriptive and associative in nature; it does not involve inferential statistical hypothesis testing, predictive modelling or causal inference. The Power BI component covers dashboard design, KPI definition and DAX measures built directly on this dataset.

Significance

This project is significant for four groups of stakeholders. For recruiters, it provides a structured, multi-metric shortlisting mechanism rather than reliance on college reputation alone. For placement cells, it provides a comparative benchmarking reference against tier, branch and state peers. For institutions and management, it identifies observed associations between institutional characteristics and placement outcomes that may inform strategic improvement. For students, it provides visibility into branch-level and skill-related employability patterns that may inform academic and career planning.

REVIEW OF CONCEPTS & LITERATURE

Engineering Education and Employability

Employability in engineering education is broadly understood as the combination of technical knowledge, practical skills, and industry readiness that enables graduates to secure and perform effectively in relevant roles. Employability is influenced by curriculum design, exposure to industry-relevant tools, internships and soft-skill development, in addition to academic performance.

Campus Placement

Campus placement functions as a structured mechanism connecting graduating students with recruiting organizations through institutional placement cells. Placement percentage is commonly used as a headline indicator of institutional performance, although it is influenced by branch composition, regional industry presence, and recruiter relationships.

Industry-Academia Skill Alignment

A recurring theme in engineering education discourse is the alignment (or misalignment) between skills taught in academic curricula and skills demanded by recruiting industries. This project operationalizes this theme directly through its Curriculum_Subjects, Skills-Taught_In_Colleges, Skills_Demanded_By_Recruiters and Missing_Skills_Gap fields.

Internship and Employability

Internship participation is widely regarded as a practical bridge between academic learning and industry expectation, offering exposure to real-world problem solving, professional environments and applied skill development. This project treats Internship_Participation_Pct as one of the three core indicators in its Recruiter Readiness framework.

Data Analytics in Recruitment

Data analytics applied to recruitment and placement functions enables organizations to move from reputation-based decision-making toward evidence-based comparison across multiple quantitative and qualitative indicators. This project applies exploratory data analysis techniques to a large, structured placement dataset for this purpose.

Business Intelligence and Power BI

Business Intelligence (BI) tools such as Power BI enable the transformation of structured datasets into interactive dashboards supporting exploratory and decision-oriented use by non-technical stakeholders. This project uses Power BI to operationalize the analytical dataset into KPI cards, comparative visuals, slicers and DAX-based readiness measures for recruiter and institutional audiences.

Research Gap

While placement percentage and salary figures for individual colleges are often available in isolation, a consolidated, cleaned, multi-dimensional dataset that jointly analyzes placement, employability, internship participation, salary, branch, tier, state, category, ownership and skill-alignment fields combined with a recruiter-oriented Power BI decision-support layer is comparatively uncommon. This project addresses that specific combination.



DATASET AND DATA SOURCES

Dataset Overview

The final analytical dataset used for this project comprises 22,386 rows and 20 columns. It was constructed after a structured cleaning and standardization process. The dataset covers 2,813 unique colleges, 33 unique states, 7 unique engineering branches, 26 unique institutional categories, 7 unique ownership types and 4 unique tiers.

Data Structure

Table lists the 20 columns retained in the final analytical dataset along with a brief functional definition of each.

#	Column	Description
1	College_Name	Name of the engineering college / institution.
2	State_Clean	Standardized state field used for analysis (33 categories).
3	Branch_Name	Engineering branch offered (7 categories).
4	Placement_Percentage	Percentage of students placed for the given college-branch record.
5	Average_Salary_LPA	Average salary offered, in Lakhs Per Annum (LPA).
6	Internship_Participation_Pct	Percentage of students who participated in internships.
7	Employability_Indicator	Composite employability score for the college-branch record.
8	State	Original, unstandardized state / location field, retained for traceability.
9	Ownership	Ownership type of the institution (e.g., Government, Private, etc.).
10	Branch_Code	Coded identifier corresponding to Branch_Name.
11	Curriculum_Subjects	Subjects taught as part of the branch curriculum.
12	Recruiter_Demand	Recorded recruiter demand information for the branch/college.
13	Industry_Skills_Required	Skills required by industry for the branch.
14	Top_Recruiters	Recorded top recruiters associated with the record.
15	Skills-Taught_In_Colleges	Skills reported as taught within the college curriculum.
16	Skills_Demanded_By_Recruiters	Skills reported as demanded by recruiters.
17	Missing_Skills_Gap	Recorded gap between taught and demanded skills.
18	Emerging_Tech_Requirements	Emerging technology requirements associated with the branch/record.

#	Column Name	Description
19	Tier	Institutional tier classification (Tier 1–4).
20	Category	Institutional category classification (26 categories).

Table — Final Dataset Structure (Columns and Definitions)

Variables

The dataset contains four core quantitative variables used throughout the descriptive and comparative analysis: Placement_Percentage, Average_Salary_LPA, Internship_Participation_Pct and Employability_Indicator. It also contains categorical/grouping variables (State_Clean, Branch_Name, Ownership, Tier, Category) and descriptive text-based fields (Curriculum_Subjects, Recruiter_Demand, Industry_Skills_Required, Top_Recruiters, Skills-Taught_In_Colleges, Skills_Demanded_By_Recruiters, Missing_Skills_Gap, Emerging_Tech_Requirements) used for skill-gap and recruiter-demand analysis.

Data Sources

The source of this dataset is official government websites like AICTE, NIRF from which this data is scraped carefully.

Data Quality

The final dataset, after the cleaning process contains zero missing values and zero duplicate analytical records, where the analytical key is defined as the combination of College_Name, State_Clean and Branch_Name. Data quality validation results are presented in detail in table.

DATA CLEANING AND PREPROCESSING

Initial Dataset Inspection

The raw dataset was first inspected to understand its shape, column names, data types and general structure prior to any transformation. This step established a baseline understanding of the scope of the cleaning task.

Column Identification

All source columns were identified and mapped against the 20 columns retained in the final analytical dataset, ensuring that every retained column had a clear, consistent definition.

Data Type Validation

Each column was validated against its expected data type - numeric fields (Placement_Percentage, Average_Salary_LPA, Internship_Participation_Pct, Employability_Indicator) were validated as numeric, and categorical/text fields were validated as string types, prior to any statistical computation.

Missing-Value Inspection

The dataset was systematically inspected for missing values across all 20 columns. The final analytical dataset contains zero missing values.

Duplicate Inspection

Duplicate inspection was carried out using the analytical key College_Name + State_Clean + Branch_Name. Repeated college/branch information existed at various points during the raw analytical process; the final analytical dataset was consolidated so that this analytical key contains zero duplicate records.

State Standardization

The original State field contained substantial inconsistency, including combined city-and-state values and a generic “Other” category. A cleaned State_Clean field was constructed to standardize these values. Through this process, state-field complexity was reduced substantially, from hundreds of raw state/location values down to 33 standardized state categories used for analysis.

It is important to note, as a matter of transparent data-quality reporting, that not every “Other” record was resolved to a specific state during this process. The project deliberately retained unresolved “Other” values rather than guessing a college's state from incomplete information. This is treated in this report as a documented data-quality limitation and, at the same time, as an example of responsible data cleaning practice - the alternative of inferring or guessing values was deliberately avoided. The original State

column was retained alongside State_Clean for traceability, so that the standardization step remains auditable.

Category Cleaning

The institutional Category field was reviewed and cleaned to consolidate inconsistent labelling into 26 standardized category values used in the final analytical dataset.

Ownership Standardization

The Ownership field was standardized into 7 consistent ownership-type categories, removing inconsistent labelling variants observed in the raw data.

Branch Standardization

Branch_Name values were standardized into 7 consistent engineering branch categories, with a corresponding Branch_Code retained for reference.

Numerical Validation

Numeric fields were validated for logical consistency percentage-based fields (Placement_Percentage, Internship_Participation_Pct, Employability_Indicator) were checked against a valid 0–100 range, and Average_Salary_LPA was checked for non-negative, plausible values.

Placement Percentage Validation

Placement_Percentage values were specifically checked for internal consistency and plausible range prior to descriptive analysis, as this field is central to the project's core business question.

Salary Validation

Average_Salary_LPA values were checked for plausibility given the observed minimum (2.88 LPA) and maximum (30.96 LPA) values reported in Chapter 5, with attention to potential extreme values discussed further in the Limitations chapter.

Analytical Grouping

Records were grouped by Tier, Branch_Name, State_Clean, Category and Ownership as needed to support the comparative analysis presented in Chapter 5, without altering the underlying record-level data.

Tier Conflict Investigation

Where a given college/branch combination showed any inconsistency in recorded Tier classification during the raw analytical process, this was investigated prior to finalizing the analytical dataset, so that the final Tier field used in analysis is internally consistent for each analytical record.

Final Analytical Record Construction

Using the analytical key College_Name + State_Clean + Branch_Name, a final, de-duplicated set of 22,386 analytical records was constructed, forming the dataset.

Final Quality Validation

The final analytical dataset was validated against the metrics summarized in Table prior to proceeding to exploratory data analysis.

Metric	Result
Rows	22,386
Columns	20
Unique Colleges	2,813
Unique States	33
Unique Branches	7
Unique Categories	26
Unique Ownership Types	7
Unique Tiers	4
Missing Values	0
Duplicate Analytical Records	0

Data Quality Validation Summary

These validation checks matter because they establish that all subsequent descriptive statistics, comparative analyses and dashboard KPIs presented in this report are computed on a dataset that is free of missing values and free of duplicate analytical records, and that all categorical groupings (state, branch, category, ownership, tier) have been standardized prior to analysis. Without this validation step, comparative figures across tier, branch or state could be distorted by inconsistent categorization or duplicated records.

EXPLORATORY DATA ANALYSIS

Descriptive Statistics

The table presents descriptive statistics for the four core quantitative indicators used throughout this project: Placement_Percentage, Average_Salary_LPA, Internship_Participation_Pct and Employability_Indicator.

Statistic	Placement_Percentage	Average_Salary_LPA	Internship_Participation_Pct	Employability_Indicator
Mean	63.131696	5.777775	54.693764	61.723251
Median	63.30	5.405833	54.516667	62.10
Std. Dev.	11.863023	2.091092	11.441015	12.673479
Minimum	34.70	2.88	27.80	31.80
Q1	54.60	4.44	46.80	52.4375
Q3	72.00	6.67	63.10	70.80
Maximum	99.20	30.96	98.00	98.80

Central tendency: The mean and median of Placement_Percentage (63.13% and 63.30% respectively) are close to one another, suggesting a broadly symmetric central distribution for this indicator. Average_Salary_LPA shows a larger gap between mean (5.78) and median (5.41), which, together with a maximum value of 30.96 far above the third quartile of 6.67, indicates a right-skewed distribution with a small number of comparatively high-salary records pulling the mean upward relative to the median.

Variability and spread: Placement_Percentage and Employability_Indicator show similar standard deviations (11.86 and 12.67 respectively), indicating comparable variability. Internship_Participation_Pct shows a somewhat narrower spread (Std = 11.44). Average_Salary_LPA shows the largest relative spread when considered against its own mean, consistent with the right-skew noted above.

Outliers: The maximum Average_Salary_LPA value of 30.96, set against a Q3 of 6.67, is consistent with the presence of high-salary outlier records.

Why the median is useful alongside the mean: Because Average_Salary_LPA in particular shows right-skew, the median offers a more robust measure of the “typical” record than the mean alone, which can be disproportionately influenced by a small number of high-salary records. Reporting both statistics together, as done throughout this report, therefore provides a more complete and less misleading picture than either statistic in isolation. No claim of statistical significance is made for any of these observations, as no formal statistical hypothesis test was performed on this distributional data.

Placement Analysis

Placement_Percentage across the full dataset ranges from a minimum of 34.70% to a maximum of 99.20%, with a mean of 63.13% and median of 63.30%. The interquartile range (Q1 = 54.60%, Q3 = 72.00%) indicates that the middle 50% of analytical records fall within roughly a 17.4 percentage-point band, reflecting meaningful but not extreme dispersion in placement outcomes across the dataset.

Salary Analysis

Average_Salary_LPA ranges from 2.88 LPA to 30.96 LPA, with a mean of 5.78 LPA and median of 5.41 LPA. The interquartile range (Q1 = 4.44, Q3 = 6.67) shows that the bulk of records cluster in a comparatively narrow band relative to the maximum value, reinforcing the right-skew noted in Section 5.1.

Internship Analysis

Internship_Participation_Pct ranges from 27.80% to 98.00%, with a mean of 54.69% and median of 54.52%, indicating a broadly symmetric distribution similar in shape to Placement_Percentage.

Employability Analysis

Employability_Indicator ranges from 31.80 to 98.80, with a mean of 61.72 and median of 62.10. This indicator shows the largest standard deviation among the four core indicators (12.67), suggesting relatively wider dispersion in employability outcomes across the dataset compared to placement and internship participation.

Tier Analysis

Table shows the distribution of analytical records across the four institutional tiers.

Tier	Records	Percentage
Tier 1	994	4.44%
Tier 2	4,480	20.01%
Tier 3	2,821	12.60%
Tier 4	14,091	62.95%

Tier	Placement Mean	Placement Median	Salary Mean	Salary Median	Internship Mean	Internship Median	Employability Mean	Employability Median
Tier 1	72.93	73.70	9.99	8.26	65.89	66.50	73.26	74.47
Tier 2	69.95	71.20	6.89	6.79	60.93	62.10	72.00	73.40
Tier 3	60.66	60.60	5.23	5.00	52.21	52.50	58.49	59.30
Tier 4	60.77	60.80	5.24	5.03	52.42	52.60	58.29	59.10

The data shows that Tier 1 institutions exhibit the strongest average performance across all four quantitative indicators placement (72.93%), salary (9.99 LPA), internship participation (65.89%) and employability (73.26). Tier 2 institutions also perform strongly, particularly on employability (72.00) and placement (69.95%), positioning Tier 2 closer to Tier 1 than to Tier 3 or Tier 4 on these two measures. Tier 3 and Tier 4 show relatively similar average quantitative performance across all four indicators, with differences between them generally under one percentage point.

This pattern is presented strictly as an observed association: Tier 1 classification is not claimed to cause stronger placement outcomes. The association may reflect a combination of institutional characteristics, industry exposure, recruiter demand, student profile and other factors that were not isolated in this analysis. It should also be noted that the record distribution across tiers is highly uneven. Tier 4 alone accounts for 62.95% of all records and this imbalance should be kept in mind when interpreting tier- level comparisons.

Branch Analysis

Table presents branch-wise performance across the four core indicators. Each of the seven branches contains 3,198 records.

Branch	Placement %	Salary(LPA)	Internship%	Employability
Computer Science Engineering	74.36	7.16	64.61	71.92
(CSE) Information Technology (IT)	72.04	6.57	62.27	69.48
Emerging Technology Domains	69.75	6.78	61.30	70.18

Branch	Placement %	Salary (LPA)	Internship %	Employability
Electronics & Communication Engineering (ECE)	63.68	5.72	55.90	62.48
Electrical Engineering (EEE)	59.06	5.28	51.53	58.35
Mechanical Engineering	53.68	4.66	45.86	52.80
Civil Engineering	49.34	4.28	41.38	46.85

Table — Branch-wise Performance (3,198 records per branch)

Ranked by placement percentage, the branch order (highest to lowest) is: CSE (74.36%) > IT (72.04%) > Emerging Technology Domains (69.75%) > ECE (63.68%) > EEE (59.06%) > Mechanical Engineering (53.68%) > Civil Engineering (49.34%). The same relative ordering holds for salary, internship participation and employability, with the minor exception that Emerging Technology Domains records a marginally higher employability indicator (70.18) than IT (69.48) despite a lower placement percentage.

This gradient is discussed as an observed pattern rather than a judgment on any branch's inherent value. Differences in industry demand, alignment between taught and demanded skills, internship availability and exposure to emerging technologies may all be associated with the observed differences between branches; no single branch is characterized in this report as “bad” or deficient in an absolute sense.

Category Analysis

The dataset spans 26 standardized institutional categories (Category). Category-wise comparative figures are intended to be explored through the Power BI dashboard filters.

Ownership Analysis

The dataset spans 7 standardized ownership types (Ownership). Ownership-wise comparative figures are intended to be explored through the Power BI dashboard filters.

Placement vs Employability

Across the dashboard analysis, Placement_Percentage and Employability_Indicator show a positive visual relationship: college-branch records with higher employability indicators tend to also show higher placement percentages. This is reported strictly as an observed visual pattern from the dashboard scatter/comparison visual.

Skill Demand Analysis

The dataset contains descriptive fields Recruiter_Demand, Industry_Skills_Required and Skills_Demanded_By_Recruiters that capture recruiter- and industry-side skill demand information at the college-branch level. These are text-based fields; a structured skill-frequency analysis would require text parsing/tokenization, as noted in Future Scope.

Skill Gap Analysis

The dataset contains a dedicated Missing_Skills_Gap field alongside Skills-Taught-In-Colleges and Skills_Demanded_By_Recruiters, enabling a structured comparison between what colleges teach and what recruiters demand.



RECRUITER READINESS FRAMEWORK

Need for Recruiter-Focused Analysis

Recruiters evaluating engineering colleges for campus hiring typically cannot rely on a single metric such as placement percentage alone, since a high placement percentage may coexist with weak internship exposure or low employability, and vice versa. This part documents a project-defined analytical framework, developed specifically for this project, to combine multiple observable indicators into a single interpretable readiness category.

Readiness Indicators

The framework uses three core indicators already validated in Placement_Percentage, Employability_Indicator and Internship_Participation_Pct. These three were selected because they jointly capture placement outcome, holistic employability and practical industry exposure, which together provide a broader picture than any single indicator.

Readiness Score

The Recruiter Readiness Score is an analytical composite concept based on Placement_Percentage, Internship_Participation_Pct and Employability_Indicator. It is explicitly a project-defined analytical construct developed for this project and is not an official industry certification, accreditation outcome or third-party rating.

Readiness Categories

Using the three core indicators, each college-branch record is classified into one of three readiness categories using the conceptual thresholds shown in Table.

Category	Condition
Recruiter Ready	Placement_Percentage ≥ 70 AND Employability_Indicator ≥ 70 AND Internship_Participation_Pct ≥ 60
Moderately Ready	Placement_Percentage ≥ 60 AND Employability_Indicator ≥ 60 AND Internship_Participation_Pct ≥ 50
Needs Improvement	All other records (does not meet the Moderately Ready thresholds)

Table - Recruiter Readiness Category Thresholds

These thresholds are project-defined analytical cut-offs chosen for this project and are explicitly not an official industry standard, accreditation benchmark or regulatory requirement. They are intended to provide a transparent, reproducible classification logic rather than an authoritative rating.

The recruiter shortlisting under this framework should combine Placement_Percentage, Employability_Indicator, Internship_Participation_Pct, Average_Salary_LPA, Branch_Name, Tier and available skill-alignment fields (Industry_Skills_Required, Skills_Demanded_By_Recruiters, Missing_Skills_Gap), rather than relying on the Readiness Category alone or on any single metric in isolation.

Interpretation

The Readiness Category should be interpreted as a starting filter for recruiter shortlisting rather than a final decision rule. A record classified as “Needs Improvement” under this framework may still be worth recruiter attention for a specific branch or skill requirement, and a record classified as “Recruiter Ready” should still be examined against branch-specific skill alignment before final shortlisting decisions are made.



POWER BI DASHBOARD DEVELOPMENT

Dashboard Objectives

The Power BI dashboard component of this project is designed around the central question: “Why are engineering college placements low, and which colleges/branches are more attractive for recruiters?” The dashboard design is deliberately recruiter-focused rather than merely descriptive, translating the analytical dataset and the Recruiter Readiness framework into an interactive decision-support tool.

Data Import

The final analytical dataset (22,386 rows × 20 columns) was imported into Power BI as the single source table underlying all dashboards described in this part. The data is imported after checking the data quality.

Data Model

The dashboards are built on a single flat analytical table rather than a multi-table star schema, consistent with the structure of the final analytical dataset as described.

Measures

The following DAX-based analytical measures underpin the dashboards. Where exact DAX formulas were available in the source material, they should be inserted verbatim by the author; otherwise, each measure is explained conceptually only, and no exact DAX syntax is fabricated here.

- Recruiter Ready Colleges -- a count measure that totals college-branch records classified as “Recruiter Ready” under the thresholds.
- Moderately Ready Colleges -- a count measure that totals records classified as “Moderately Ready.”
- Needs Improvement Colleges -- a count measure that totals records classified as “Needs Improvement.”
- Recruiter Readiness Score -- an analytical composite concept based on Placement_Percentage, Internship_Participation_Pct and Employability_Indicator, etc.

These calculated measures are explicitly distinguished from the raw dataset columns from which they are derived.

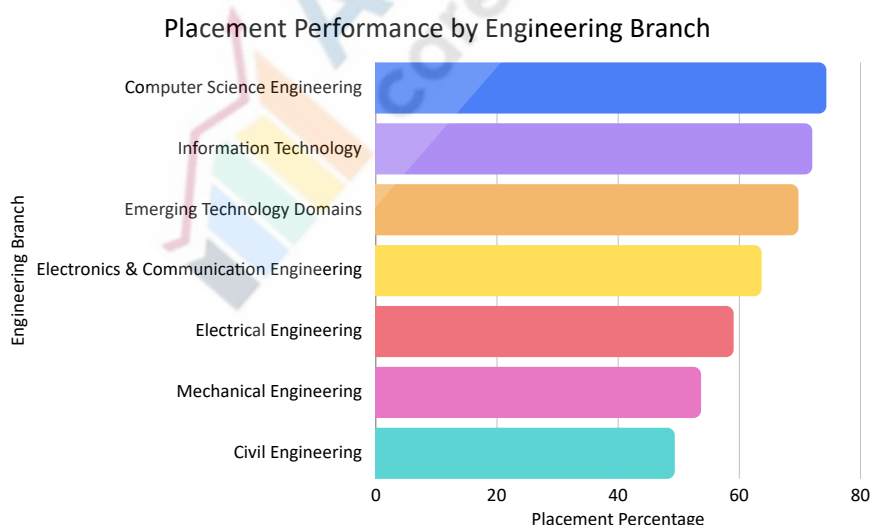
Dashboard 1 - Engineering College Placement & Recruitment Intelligence

Purpose: This dashboard provides an overview of the full dataset, oriented toward understanding aggregate and comparative placement performance across branches and tiers.

Element	Detail
KPI Cards	Engineering Colleges; Average Placement %; Average Salary (LPA); Internship Participation %; Employability Index; Analytical Records
Core Visuals	Branch placement performance; Employability vs Placement; Tier comparison; Branch/Tier comparisons
Filters / Slicers	Available for interactive filtering across the dashboard Tier, Category, Ownership, State_Clean, Branch_Name
Intended Audience	Institutional management, placement cells, and recruiters seeking an overall view
Business Questions Answered	What is overall placement/salary/internship/employability performance? How do branches and tiers compare?

Table - Dashboard 1 - KPI Cards and Visuals

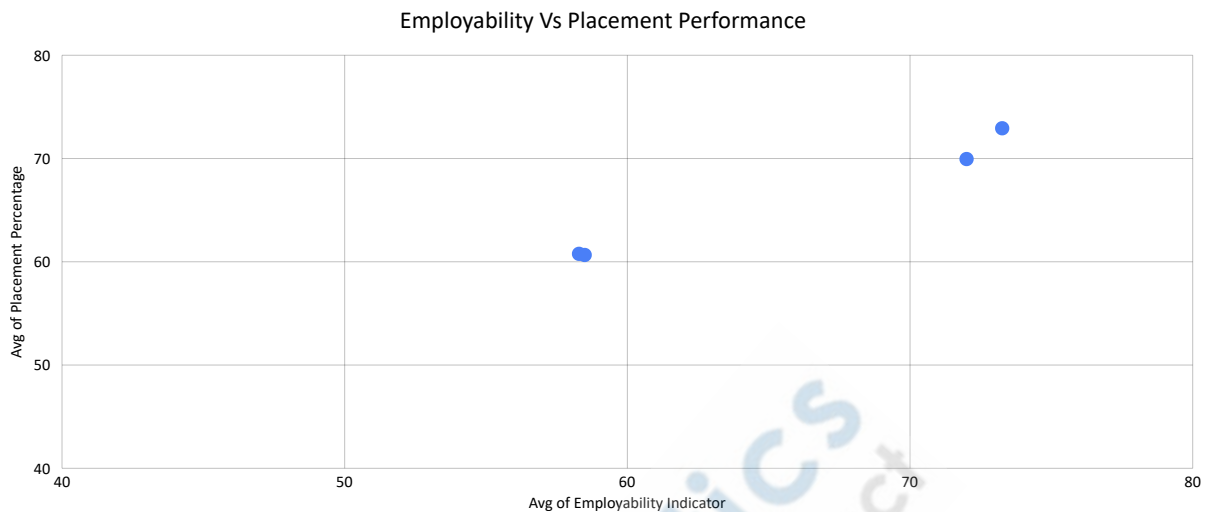
Important Charts



Interpretation: Computer Science Engineering has the strongest placement performance, with an average placement rate of 74.36%.

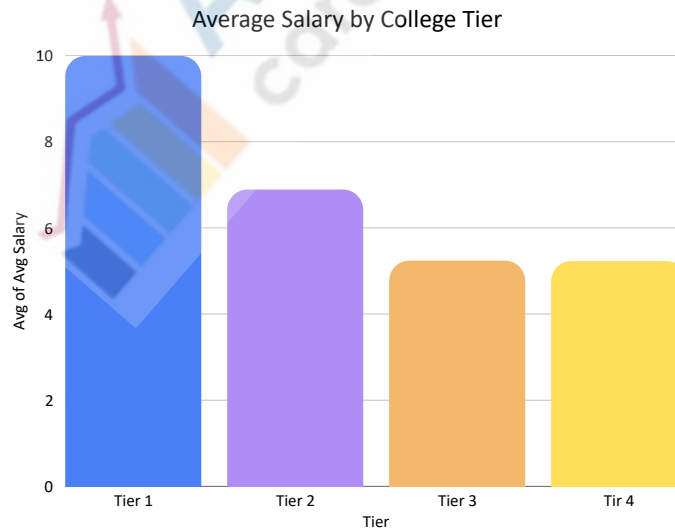
Information Technology follows closely at 72.04%, while Emerging Technology domains achieve 69.75%.

At the lower end, Civil Engineering has an average placement rate of only 49.34%, followed by Mechanical Engineering at 53.68%.



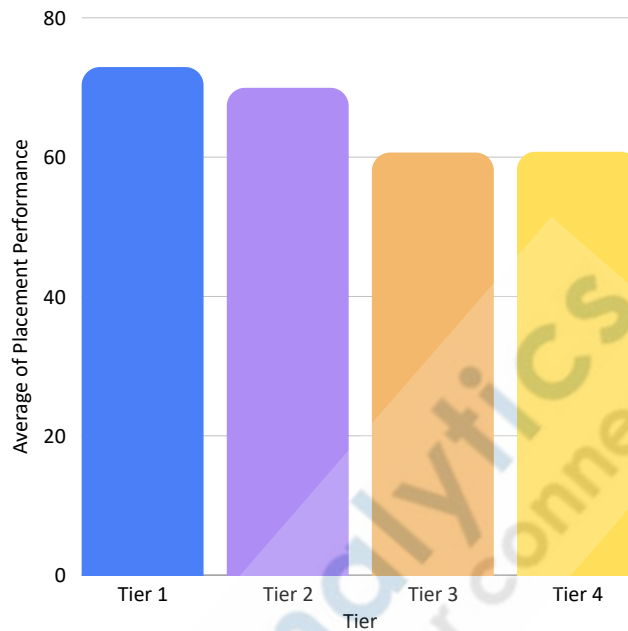
Interpretation: The Scatterplot shows a positive relationship between employability and placement performance.

This shows higher employability is associated with stronger placement outcomes. It suggests that placement problems should not be examined only from the perspective of the number of recruiters visiting colleges. They should also be examined through employability, internships, industry skills, curriculum relevance, recruiter demand, skill gap, etc.



Interpretation: The salary difference between Tier 1 and Tier 4 is approximately 4.75 LPA. This suggests that college tier is associated not only with the probability of placement, but also with the quality/value of placement opportunities, as represented by salary.

Average Placement Performance by College Tier



Interpretation: Tier 1 colleges demonstrate the highest average placement performance at 72.93%, followed by Tier 2 at 69.95%. Tier 3 and Tier 4 colleges are substantially lower, at approximately 61%.

This indicates a considerable performance gap between higher-tier and lower-tier institutions.

Branch x Tier Performance Matrix

Branch_Name	Tier 1	Tier 2	Tier 3	Tier 4
Mechanical Engineering	63.88	61.33	50.75	51.12
Information Technology (IT)	80.39	77.48	70.02	70.12
Emerging Technology Domains	79.04	76.26	67.90	67.39
Electronics & Communication Engineering (ECE)	74.64	70.80	60.74	61.24
Electrical Engineering (EEE)	69.44	66.51	56.77	56.42
Computer Science Engineering (CSE)	83.40	80.13	71.97	72.37
Civil Engineering	59.69	57.12	46.47	46.71

Interpretation: The branch effect remains visible even within different college tiers.

CSE maintains relatively strong placement performance across all tiers, whereas Civil remains substantially lower. This suggests that, Placement performance is influenced by both institutional tier and engineering branch.

Dashboard 2 - Recruiter Readiness & Shortlisting Intelligence

Purpose: To answer which factors indicate recruiter readiness, what skills are missing, and which engineering branches present stronger recruitment potential?

Element	Detail
KPI Cards	Average Placement %; Average Salary; Average Employability; Recruiter Ready Colleges; Internship Participation
Core Visuals	Recruiter demand by Branch; Placement VS Employability; Top Skills missing among engineering colleges
Filters / Slicers	Tier
Intended Audience	Recruiters performing college/branch shortlisting
Business Questions Answered	Which engineering branch has more recruiter demand? Are placement & employability dependent? What are the top skills missing among engineering colleges?

Table - Dashboard 2 - KPI Cards and Visuals

Important Charts & KPIs

Average Employability = 61.72

Interpretation: The overall employability indicator was 61.72, suggesting moderate industry readiness across the engineering-college dataset and indicating scope for strengthening employment-oriented skills and competencies.

Average Placement = 63.13%

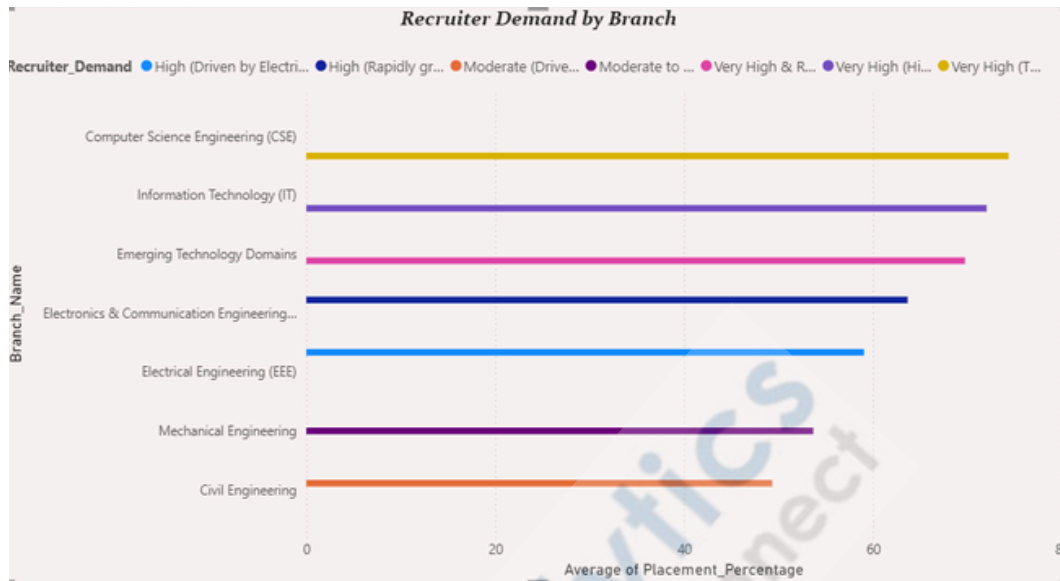
Interpretation: This means the dataset's overall placement performance is considerably below a highly successful placement environment. This allows us to examine placement as a multi-dimensional recruitment outcome, rather than looking only at placement percentage.

Recruiter-Ready Colleges $\approx$ 1K

Interpretation: This indicates that a substantial subset of colleges meets defined recruiter-readiness criteria. These colleges represent potential high-priority institutions for recruiters under project's defined readiness criteria.

Internship Participation = 54.69%

Interpretation: The overall internship participation rate is 54.69%, indicating that internship participation is moderate across the engineering-college records. Approximately half of the represented student records is associated with internship participation, suggesting considerable scope for increasing practical industry exposure.



Interpretation: Technology-oriented branches appear to have stronger recruiter-oriented outcomes, while traditional engineering branches such as Mechanical and Civil show comparatively weaker placement performance.

Top missing skills among Engineering Colleges(Bar Chart)

The chart shows numerous industry-oriented skills appearing as missing skills, including areas such as:

- Advanced FEA/CFD Simulation
- Advanced Structural Software(ETABS)
- Advanced VLSI Physical Design
- Agile Workflow
- Applied Cybersecurity Operations
- Automated CNC Programming
- BIM 3D/4D Integration
- Cloud Security Frameworks
- Digital Project management (Primavera/MS Project)
- Drone Surveying
- EV Powertrain Design, etc.

Interpretation: The missing-skill data suggests that engineering colleges are not facing a single universal skill problem.

Instead, the gaps are domain-specific.

The recruiter-readiness analysis indicates that engineering-college placement outcomes are associated with multiple dimensions of workforce readiness. Higher employability levels correspond with stronger placement performance, while substantial differences are observed across engineering branches and institutional tiers. The analysis also highlights domain-specific skill gaps, particularly in emerging technologies, advanced engineering tools, cloud technologies, automation and industry-specific applications. These findings suggest that improving placement outcomes requires stronger alignment between curriculum, industry skill requirements, practical exposure and recruiter demand.

Dashboard 3 - Recruiter Shortlisting & College Comparison

Purpose: To identify high-performing colleges and branches by comparing placement performance, salary, employability, and recruiter readiness across college tiers.

Element	Detail
KPI Cards	Total Colleges, Average Placement %; Average Salary; Average Employability; Recruiter Ready Colleges; Moderately Ready Colleges; Needs Improvement Colleges
Core Visuals	College Placement VS Employability; Colleges by Placement Performance; College Comparison Table
Filters / Slicers	Tier
Intended Audience	Recruiters performing college/branch shortlisting
Business Questions Answered	Which engineering college placements leads more employability? Which are top colleges by placement performance?

Table - Dashboard 3 - KPI Cards and Visuals

Important Charts & KPIs

Colleges Available — ~2,811

Interpretation: The dataset covers around 2,811 colleges, giving a broad view of engineering-college placement and employability performance. This makes the analysis useful for comparing institutions across different tiers, states, branches, and ownership types.

Average Placement — 63.13%

Interpretation: On average, around 63% of students are placed across the colleges in the dataset. This indicates that placement outcomes are moderate overall, but there is still considerable room for improvement across institutions.

Average Salary — 5.78 LPA

Interpretation: The overall average salary is approximately ₹5.78 LPA. This provides a useful benchmark for comparing colleges and branches, particularly when recruiters are evaluating potential talent pools.

Average Employability — 61.72

Interpretation: The average employability indicator is 61.72, which is broadly aligned with the overall placement level of 63.13%. This suggests that employability and actual placement outcomes are relatively closely connected in the dataset.

Recruiter Ready Colleges — 416

Interpretation: Around 416 colleges fall into the dashboard's Recruiter Ready category based on the project's readiness criteria.

Moderately Ready Colleges — 482

Interpretation: Another 482 colleges fall into the Moderately Ready category. These institutions show reasonable potential but may not perform as consistently as the recruiter-ready group.

Needs Improvement Colleges — ~2K

Interpretation: The largest group is the Needs Improvement category, containing roughly 2,000 colleges. This is one of the most important findings on the dashboard because it shows that the majority of institutions do not currently meet the project's threshold for strong recruiter readiness.

College Placement vs Employability — Scatter Plot

Interpretation: There is a clear positive relationship visible between placement and employability. The higher-placement colleges generally tend to have higher employability scores as well. The Tier 1 and Tier 2 clusters are concentrated toward the higher-performing region, while Tier 3 and particularly Tier 4 colleges are more concentrated around lower placement and employability levels. There are also some individual colleges that perform better than the general pattern for their tier.

Colleges by Placement Performance — Bar Chart

Interpretation: This visual highlights colleges with comparatively strong placement performance. The leading colleges shown in the current view have placement performance toward the higher end of the dataset.

The chart helps recruiters quickly identify institutions that can potentially be prioritized for campus hiring. Instead of looking through thousands of colleges individually, recruiters can use this type of ranking to identify high-performing institutions first.

College Comparison Table

The College Comparison Table is designed to help recruiters compare placement performance across engineering branches and college tiers. It provides a quick view of how the same branch performs across Tier 1–Tier 4 institutions.

Interpretation: The college comparison analysis reveals substantial variation in placement performance across both engineering branches and institutional tiers. CSE, IT and Emerging Technology domains demonstrate comparatively stronger placement outcomes across tiers, whereas Civil and Mechanical Engineering show lower performance, particularly in Tier 3 and Tier 4 institutions. The results suggest that recruiter shortlisting can be made more effective by considering branch-specific performance alongside institutional tier rather than using college tier as the sole selection criterion.

Dashboard Conclusion

- Placement performance varies significantly across college tiers, with Tier 1 colleges generally showing stronger outcomes.
- Many Tier 3 and Tier 4 colleges have moderate placement but comparatively lower employability, highlighting a job-readiness gap.
- Salary outcomes also differ across colleges, suggesting that placement alone does not guarantee strong career opportunities.
- A large number of colleges fall into the “Needs Improvement” category, indicating gaps in skills, industry readiness, or practical exposure.
- Recruiter-ready colleges are concentrated among stronger-performing institutions, while lower-tier colleges need better alignment with industry requirements.
- Overall, the dashboard supports our problem statement that students can complete college and still lack the skills and employability needed to become job-ready.

Dashboard 4 - State & Institutional Performance Analysis

Purpose: To identify regional and institutional factors affecting engineering college placements, by comparing states, ownership types, institution categories, employability, salaries, and placement performance.

Element	Detail
KPI Cards	States Covered, Average Placement %; Average Salary; Average Employability; Average Internship participation
Core Visuals	Average graduate salary by State(Bar Chart), Placement performance(Column Chart), State Placement Vs Employability(Scatter Plot), Employee College Placement VS Employability; Colleges by Placement Performance;
Filters / Slicers	Tier, State_Clean, Branch_Name
Intended Audience	Recruiters performing college/branch shortlisting
Business Questions Answered	Which colleges/branches are Recruiter Ready? How do colleges compare on placement vs employability? How many institutions fall into each readiness category, after applying tier/state/branch/category/ownership filters?

Table - Dashboard 4 - KPI Cards and Visuals

KPI Cards

33 states covered -- The dataset provides broad geographical coverage.

Average Placement: 63.13% -- Overall placement performance is moderate.

Average Salary: 5.78 LPA – Represents the overall average graduate salary.

Average Employability: 61.72 -- Indicates moderate overall employability.

Average Internship Participation: 54.69% -- Around half of the records show internship participation, suggesting scope for stronger industry exposure.

Average Graduate Salary by State - Bar Chart

Salary varies considerably across states.

Manipur, Puducherry, Mizoram and Assam appear among the higher-performing states in the displayed ranking.

This indicates that geographical location alone does not completely determine salary outcomes; institutional and branch-level factors may also contribute.

Placement Performance by Ownership - Column Chart

Placement performance differs across ownership types.

Government/aided and autonomous institutions generally show relatively stronger placement outcomes than some private categories.

This suggests that institutional structure and ownership may be relevant factors when evaluating placement performance.

Average Salary by Institution Category - Bar Chart

IIIT, IIT and NIT appear among the higher salary-performing institution categories.

This highlights the strong salary outcomes associated with premier technical institutions.

Recruiters can therefore use institution category as one factor in identifying potentially high-value talent pools.

Placement Performance by Institution Category - Column Chart

Placement percentages vary across institution categories, with several premier/autonomous categories showing comparatively strong outcomes.

The visual reinforces that institutional category is associated with differences in placement performance.

State Placement vs Employability - Scatter Plot

The scatter plot shows a positive relationship between placement percentage and employability.

States/institutional groups with higher employability generally tend to have higher placement performance.

This supports the project's central finding that employability is an important dimension of placement outcomes.

Recruiter Opportunity: Employability vs Placement - Scatter Plot

The state-level scatter plot shows a clear upward trend between employability and placement.

This is particularly useful for recruiters because it indicates that areas with stronger employability indicators may provide stronger recruitment opportunities.

The relationship should be interpreted as an association, not proof of causation.

Average Placement Performance by State - Bar Chart

Placement performance differs substantially between states.

The displayed ranking highlights Manipur, Mizoram, Puducherry and Chandigarh among the higher-performing states.

This suggests that recruiters can consider regional talent pools rather than concentrating recruitment only in traditionally dominant locations.

Category–Tier Comparison Table

This table enables comparison of placement and salary performance by institution category within a selected tier.

It is useful for identifying which institution categories perform better within the same tier.

For recruiters, this provides a more granular shortlisting approach than looking at tier alone.

Dashboard Conclusion: Dashboard 4 demonstrates that engineering placement outcomes vary across states, ownership types and institution categories, while employability shows a positive association with placement performance. The findings suggest that recruiters can improve talent sourcing by considering regional and institutional differences rather than relying solely on college tier.

RESULTS AND FINDINGS

Overall Findings

The results indicate that engineering-college placement performance is moderate but highly heterogeneous across branches, college tiers, institutional categories, ownership types and geographical regions.

The overall average placement percentage is 63.13%, with a median of 63.30%. The average graduate salary is 5.78 LPA, with a median of 5.41 LPA. Internship participation averages 54.69%, while the average Employability Indicator is 61.72.

The relatively close mean and median values for placement indicate that placement performance is centered around approximately 63%. In contrast, salary shows greater positive skew because the maximum salary reaches 30.96 LPA, considerably above the median.

Branch Findings

Significant differences are observed among the seven engineering branches.

- Computer Science Engineering (CSE) records the highest average placement percentage at 74.36%, followed by Information Technology (IT) at 72.04% and Emerging Technology Domains at 69.75%.
- Electronics & Communication Engineering (ECE) records an average placement of 63.68%, while Electrical Engineering (EEE) records 59.06%.
- Mechanical Engineering records 53.68%, while Civil Engineering records the lowest average placement at 49.34%.

A similar pattern is observed for salary and internship participation. CSE records the highest average salary at 7.16 LPA and the highest internship participation at 64.61%. Civil Engineering records the lowest values for both salary (4.28 LPA) and internship participation (41.38%).

The overall branch performance gradient is:

CSE > IT > Emerging Technology Domains > ECE > EEE > Mechanical Engineering > Civil Engineering

This consistent pattern across placement, salary and internship participation indicates substantial differences in recruitment outcomes among engineering disciplines.

From a recruiter perspective, technology-oriented branches provide comparatively stronger placement and employability outcomes, whereas traditional branches, particularly Civil and Mechanical Engineering, represent areas where additional industry alignment and employability development may be required.

Tier Findings

College tier is strongly associated with differences in placement outcomes.

Tier 1 institutions record the strongest average performance:

- Placement: 72.93%
- Salary: 9.99 LPA
- Internship participation: 65.89%
- Employability: 73.26

Tier 2 institutions also demonstrate strong performance, with:

- Placement: 69.95%
- Salary: 6.89 LPA
- Internship participation: 60.93%
- Employability: 72.00

Tier 3 institutions record:

- Placement: 60.66%
- Salary: 5.23 LPA
- Internship participation: 52.21%
- Employability: 58.49

Tier 4 institutions record:

- Placement: 60.77%
- Salary: 5.24 LPA
- Internship participation: 52.42%
- Employability: 58.29

An important finding is that Tier 3 and Tier 4 show very similar quantitative performance, particularly for placement, salary and internship participation.

The dataset is heavily weighted toward Tier 4, which represents approximately 62.95% of all analytical records. Therefore, tier comparisons should be interpreted together with the distribution of records rather than assuming that the overall dataset is equally represented across tiers.

The results demonstrate that institutional tier is an important differentiating factor, but it should not be treated as the only criterion for recruiter shortlisting.

State Findings

- The analysis covers 33 standardized states/regions, enabling geographical comparison of engineering-college performance.
- The Power BI analysis demonstrates variation in placement and salary outcomes across states. State-level salary and placement rankings show that some regions perform considerably better than others, indicating that recruitment opportunities are not geographically uniform.
- The state-level placement versus employability analysis also demonstrates a positive visual relationship between Employability Indicator and Placement Percentage. States or state-level groups with higher employability values generally tend to show stronger placement outcomes.
- This finding supports the importance of employability development in improving recruitment outcomes while also demonstrating that recruiters can use geographical analysis to identify potentially strong regional talent pools.
- However, state-level results should be interpreted alongside college tier, branch and institutional category because differences in the composition of institutions across states can influence aggregate state performance.

Employability Findings

- The Employability Indicator has a mean of 61.72 and a median of 62.10, with values ranging from 31.80 to 98.80.
- Among the four core quantitative indicators, Employability Indicator has the highest standard deviation at approximately 12.67, indicating substantial variation in employability outcomes across the dataset.
- A positive relationship is observed between placement percentage and employability in the Power BI scatter-plot analysis. Colleges or groups with higher employability levels generally tend to demonstrate higher placement performance.
- Branch-level results further reinforce this relationship. CSE records an average employability of 71.92, while Civil Engineering records the lowest average of 46.85.
- Tier-level results show the same pattern, with Tier 1 recording an average employability of 73.26, compared with 58.29 for Tier 4.
- Therefore, employability emerges as one of the most important dimensions associated with placement performance in this analysis.
- The relationship should be interpreted as an association rather than evidence of causation.

Internship Findings

- The Internship Participation Percentage ranges from 27.80% to 98.00%, with an overall mean of 54.69% and median of 54.52%.
- Internship participation varies substantially across engineering branches and college tiers.
- CSE records the highest branch-level internship participation at 64.61%, followed by IT at 62.27%. Civil Engineering records the lowest participation at 41.38%.
- Similarly, Tier 1 institutions record the highest internship participation at 65.89%, followed by Tier 2 at 60.93%. Tier 3 and Tier 4 record substantially lower values of 52.21% and 52.42%, respectively.

- The results show that higher internship participation generally occurs alongside stronger placement, salary and employability outcomes.
- This suggests that industry exposure is an important component of the broader recruiter-readiness framework. However, internship participation should be considered an associated indicator rather than interpreted as a direct causal determinant of placement.

Salary Findings

- The overall average salary is 5.78 LPA, with a median of 5.41 LPA.
- Salary varies substantially according to branch and college tier.
- At the branch level, CSE records the highest average salary at 7.16 LPA, followed by Emerging Technology Domains at 6.78 LPA and IT at 6.57 LPA.
- Civil Engineering records the lowest average salary at 4.28 LPA, followed by Mechanical Engineering at 4.66 LPA.
- At the tier level, Tier 1 records the highest average salary at 9.99 LPA, followed by Tier 2 at 6.89 LPA. Tier 3 and Tier 4 are closely aligned at approximately 5.23–5.24 LPA.
- The maximum observed salary of 30.96 LPA, compared with the median of 5.41 LPA, indicates the presence of high-value salary observations and a right-skewed salary distribution.
- The results therefore indicate that salary outcomes are strongly differentiated by both institutional tier and engineering discipline.

Skill Gap Findings

The descriptive fields in the dataset provide an important industry-alignment dimension through:

- Curriculum Subjects
- Industry Skills Required
- Skills Taught in Colleges
- Skills Demanded by Recruiters
- Missing Skills Gap
- Emerging Technology Requirements
- Recruiter Demand

The Power BI skill-gap analysis identifies recurring missing-skill areas across engineering colleges. The dashboard highlights gaps related to advanced technical and industry-oriented capabilities, including areas such as advanced FEA, advanced structural technologies, advanced VLSI, Agile workflows, cybersecurity, BIM/3D–4D integration, cloud security, EV powertrain technologies, vehicle dynamics, and firmware/RTOS-related skills.

These findings indicate that the placement problem is not solely an issue of academic qualification. There is also a potential industry-readiness and skill-alignment gap between capabilities developed in colleges and those demanded by recruiters.

The skill-gap findings should consequently be interpreted as evidence of recurring industry-skill requirements and potential areas for curriculum and training improvement, rather than as causal evidence of why an individual student was not placed.

Recruiter Shortlisting Findings

The recruiter-readiness analysis combines the major quantitative indicators of:

- Placement performance
- Internship participation
- Employability

to classify colleges into different levels of recruiter readiness.

The Power BI dashboard provides three major categories:

1. Recruiter Ready
2. Moderately Ready
3. Needs Improvement

The dashboard enables recruiters to filter these categories by college tier, state, engineering branch, institutional category and ownership.

The analysis therefore moves beyond simply ranking colleges by placement percentage. It provides a multidimensional approach to identifying institutions that demonstrate stronger combinations of placement performance, employability and industry exposure.

The results also show that a large proportion of the college population falls outside the highest-readiness category under the defined thresholds. This indicates substantial scope for improving recruiter readiness across the engineering-college ecosystem.

For recruiters, the key finding is that college tier alone should not determine shortlisting. Combining placement, employability and internship indicators allows recruiters to identify potentially strong institutions across different tiers and regions.

Integrated Findings from the Power BI Dashboards

- Placement performance varies considerably across branches, tiers, states and institutional characteristics.
- CSE and IT perform strongly, showing higher placement, salary, internship participation and employability.
- Tier 1 institutions lead overall, but college tier alone is not sufficient for recruiter shortlisting.
- Higher employability is associated with higher placement performance, highlighting its importance for recruitment outcomes.
- Internship participation is higher among stronger-performing branches and tiers, indicating the importance of industry exposure.
- Significant skill gaps exist, particularly in advanced technical and industry-oriented skills demanded by recruiters.

- Multidimensional shortlisting is more effective, combining placement, employability, internships, branch, tier, state and skill requirements.
- Civil and Mechanical Engineering show weaker outcomes, highlighting opportunities for improved industry exposure, practical training and skill development.



CONCLUSION

The study shows that engineering-college placement performance is influenced by a combination of branch, institutional tier, employability, internship exposure, geographical factors and industry-skill alignment. The overall placement rate of 63.13% indicates that a considerable proportion of students still remain outside placement opportunities.

The analysis clearly shows that CSE and IT have stronger placement, salary, internship and employability outcomes, while Civil and Mechanical Engineering show comparatively weaker performance. Similarly, Tier 1 institutions perform better overall, although the analysis also shows that college tier alone should not be used to judge recruitment potential.

One of the important findings is the positive relationship between employability and placement performance. The internship analysis also indicates that institutions with stronger industry exposure generally show better outcomes. At the same time, the skill-gap analysis highlights the need to improve alignment between college-taught skills and current recruiter requirements.

Overall, the project demonstrates that the engineering placement problem is not caused by a single factor. A combination of better industry exposure, practical skill development, employability enhancement and curriculum alignment is required to improve placement outcomes. The Power BI dashboards further demonstrate how these factors can be converted into practical insights for recruiter shortlisting, college comparison and institutional improvement.

RECOMMENDATIONS

- **Strengthen industry-oriented skills**

Colleges should regularly update technical training according to the skills currently demanded by recruiters.

- **Increase internship opportunities**

More internships, live projects and industry interactions should be provided, particularly in lower-performing branches and institutions.

- **Improve employability skills**

Colleges should focus on problem-solving, communication, aptitude, technical interviews and practical project experience along with academic knowledge.

- **Reduce branch-level skill gaps**

Traditional branches such as Civil and Mechanical Engineering can benefit from additional training in emerging technologies and industry-specific tools.

- **Use data-driven recruiter shortlisting**

Recruiters should consider placement, employability, internship participation, branch and skill alignment together rather than relying only on college tier or reputation.

- **Regularly monitor placement performance**

Institutions should track placement, salary, internship and employability indicators over time to identify areas requiring intervention.

- **Build stronger industry-academia collaboration**

Colleges and recruiters should work together through industry projects, curriculum inputs, workshops, internships and placement-oriented training.

- **Use dashboards for continuous decision-making**

The developed Power BI dashboards can be used by colleges and recruiters to compare institutions, identify skill gaps and support evidence-based recruitment decisions.

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